

K774 — Working Target 4 + Lane B: the code frame TIGHTENS the ceiling (the top mass, not 174), but a quantization check shows “mass = coverage = n/q ” is TOP-ONLY. The light fermions are off-lattice corrections — Target 1 (full spectrum as RS code) is not the naive version.

Keeper | 2026-07-20 | Lead working the ceiling-as-capacity (Target 4) and the neutrino overflow (Lane B) while the team runs 07-20b. One pretty result, one disciplining check. `code_ceiling_quantization.py`.

Target 4a — the code frame gives a TIGHTER ceiling (conditional, LEAD)

Two levels of ceiling: - C-S (kinematic): $y \leq 1 \rightarrow m \leq v/\sqrt{2} = 174.1$ GeV. DERIVED (FA#7). - CODE (if $y = \text{coverage} = n/q$, and the maximal primitive RS block length is $n = q-1$): $y \leq (q-1)/q = 127/128 \rightarrow m \leq 172.7$ GeV. And 172.7 GeV is the top mass.

So under the code frame, FA#7 *tightens*: not “no fermion above 174” but “no fermion above the top” — because the top saturates the *maximal RS coverage* $(q-1)/q$, and nothing can cover more of a q -symbol field than $q-1$ positions. This is an information-theoretic sharpening of the falsifier. Tier: conditional on the code frame = LEAD (guards 1, 2 of 127/128 still stand; this doesn’t clear them, it re-expresses the same lead as a tighter ceiling).

Target 4b — the DISCIPLINE CHECK (this is the valuable part): coverage is TOP-ONLY

The code frame makes a *second*, testable prediction: if $y = \text{coverage} = n/q$, then every y is an integer multiple of $1/q = 1/128$. Checked all 9 fermions ($y \cdot 128$): - top: 127.01 — an integer (127). $\square$ - bottom 3.07, tau 1.31, charm 0.93, muon 0.078, ... — all non-integer, none on the lattice.

Verdict: only the top lands on the code lattice. So “mass = coverage = n/q ” is a LEADING-order (top-only) statement. The light fermions are off-lattice corrections — which is exactly consistent with the rank-1 picture (the top is the leading codeword; everyone else is a Tier-2 off-rank-1 correction), and it is *not* consistent with a naive “all fermions are RS codewords with $y = n/q$.”

Consequence for Target 1 (full spectrum as the RS Ladder): do NOT pursue “ $y_f = n_f/q$ for every fermion” — it fails for everything but the top. The full spectrum being the RS *Ladder* must be a subtler structure (the corrections to the leading codeword — the code’s error/redundancy structure, or a different code variable), not naive coverage. This is a guardrail for the team’s Target 1, saving a dead route (the naive one) before it’s chased.

Lane B — the neutrino overflow as a zero-mode (structural, LEAD)

The overflow symbol (the one field position the top’s codeword does NOT cover) has, by construction, zero overlap with the coverage — so a mode living there has zero Yukawa $\rightarrow$ massless. That is the neutrino-as-zero-mode. Whether it coincides with the rank-2 zero eigenvalue (F589’s $m_1 = 0 =$ the kernel of the measure-overlap operator) is the open identification: both are “the mode orthogonal to the condensate coverage,” so they *plausibly* coincide (two routes to one massless ν), but the explicit identification (overflow position = rank-2 kernel) is the step to prove. Tier: LEAD — the structural argument is clean; the two-route convergence needs the explicit identification.

Honest net

- Sharpening (lead): the code ceiling is the top mass — FA#7 tightens to “nothing above the top,” conditional on the code frame.
- Guardrail (the useful bit): the naive “mass = coverage” is top-only; the light fermions are off-lattice, so Target 1’s full-spectrum-as-code must be subtler than coverage. Saves a dead route.
- Lane B: overflow $\rightarrow$ zero-mode is clean; = rank-2 kernel is the open identification.

- Unchanged: 127/128 stays LEAD (guards 1, 2); angular = 1 stays derived; the code frame stays a FRAME.
- Keeper K774, 2026-07-20. Code ceiling $y \leq (q-1)/q = 127/128 \rightarrow m \leq 172.7$ = the top mass (tightens FA#7 to “nothing above the top”, conditional/LEAD). Quantization check: only the top is on the $1/q$ lattice (127); light fermions off-lattice $\rightarrow$ “mass = coverage = n/q ” is TOP-ONLY, guardrail for Target 1 (full spectrum needs a subtler code structure, not naive coverage). Lane B: overflow $\rightarrow$ zero-mode (clean) = rank-2 kernel (open identification). See [[Keeper_K773_BST_is_a_holographic_RS_error_correcting_code_synthesis_2026-07-20]].